

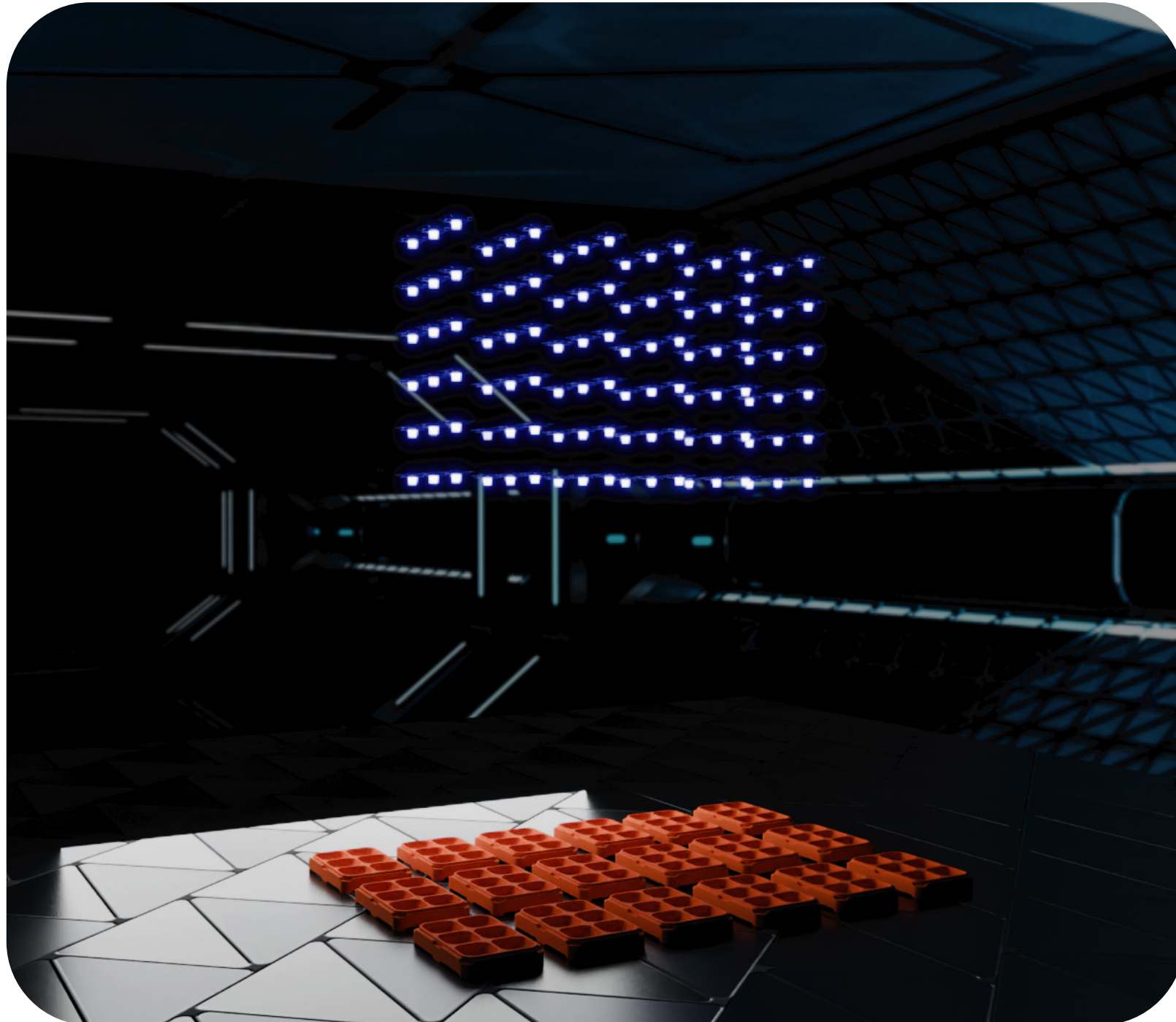
A PROPRIETARY ECOSYSTEM FOR LARGE-SCALE AERIAL ENTERTAINMENT



FLYSTACK

Advanced Drone Show Technology & Integrated Multimedia Systems

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Our mission is to design and operate safety-critical aerial systems.

We are FLYSTACK — an internationally recognized technology company operating at the intersection of aerospace engineering, software development, and large-scale multimedia drone shows.

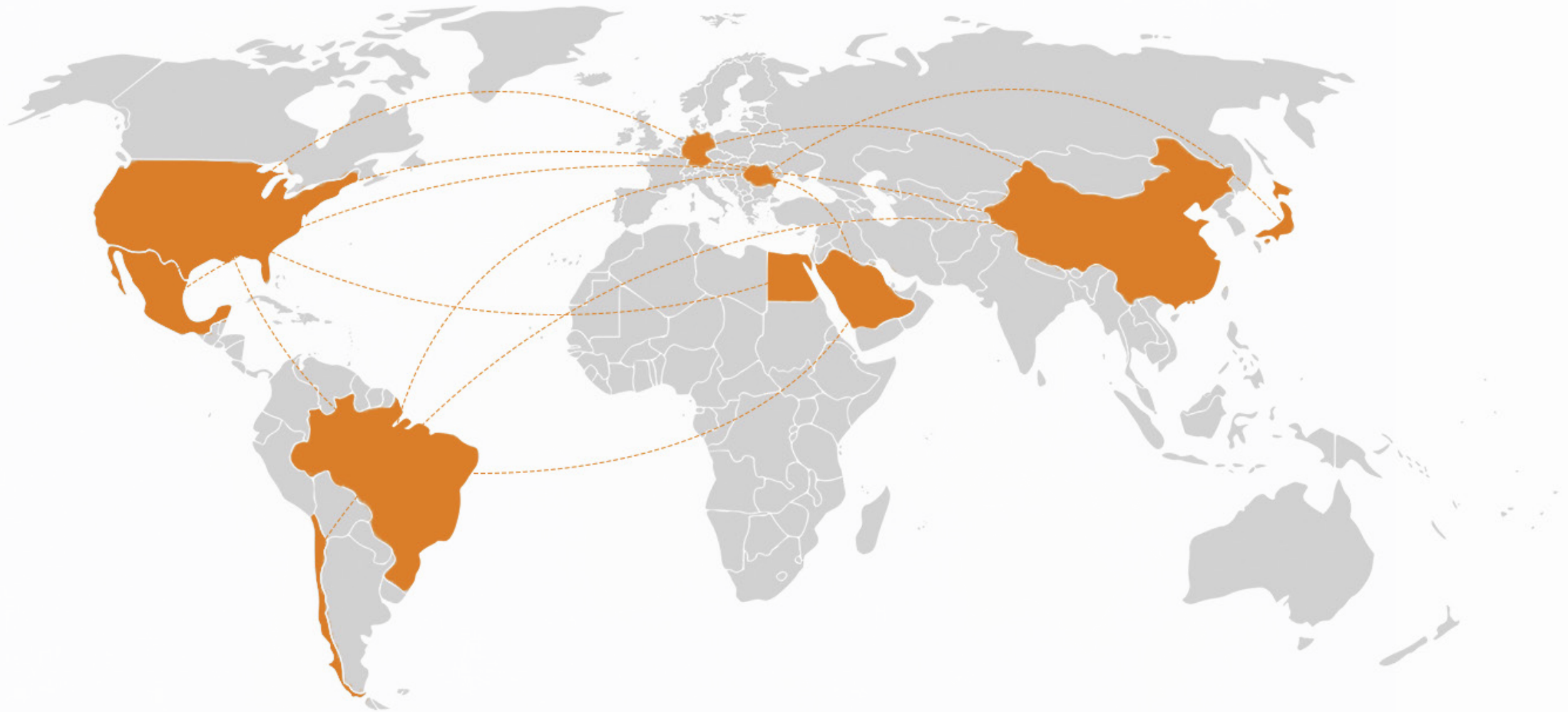
For several years, we have been actively shaping both the drone technology market and the aerial entertainment industry, delivering innovative, reliable, and scalable solutions worldwide.



To learn more, visit us at www.flystackdroneshow.com

Operating Worldwide

Today, we are present across Europe, Asia, and America, working with major brands, public institutions, and event producers on projects that redefine what is possible in the sky.



Why Large-Scale Drone Shows Are Complex?

Modern drone shows require far more than synchronized flight. Our drones are designed and produced using industrial-grade components and tested in real-world, high-complexity environments.

● Large-Scale Drone demand:

Centimeter-level positioning accuracy



Multi-layered redundancy



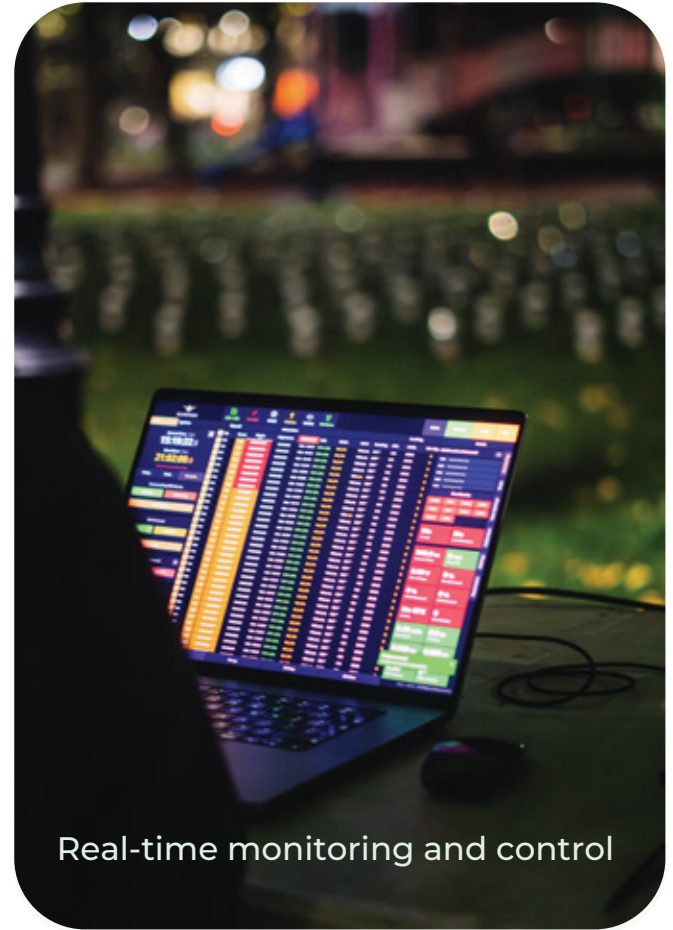
Seamless synchronization with music, pyrotechnics, lasers, and multimedia systems



Compliance with aviation regulations across different jurisdictions



Real-time monitoring and control



A Fully Integrated In-House Ecosystem

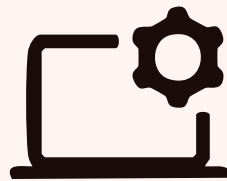
FLYSTACK designs, develops, and operates every critical layer of a drone show production. This full-stack control enables unmatched safety, precision, and scalability - from hundreds to thousands of drones.

Hardware



Proprietary drone hardware

Software



Custom-built control and animation software

Operations & Shows



Creative, technical, and operational expertise



HARDWARE PRODUCTS

Drone Platforms Engineered for Professional Aerial
Entertainment



NOVA

NOVA features case-based takeoff and landing for fast, pad-free deployment, high-output 32W LEDs delivering up to 1000 lumens for long-distance visibility, and a rain-proof design built for outdoor operations.

NOVA

NOVA represents the future of drone show technology

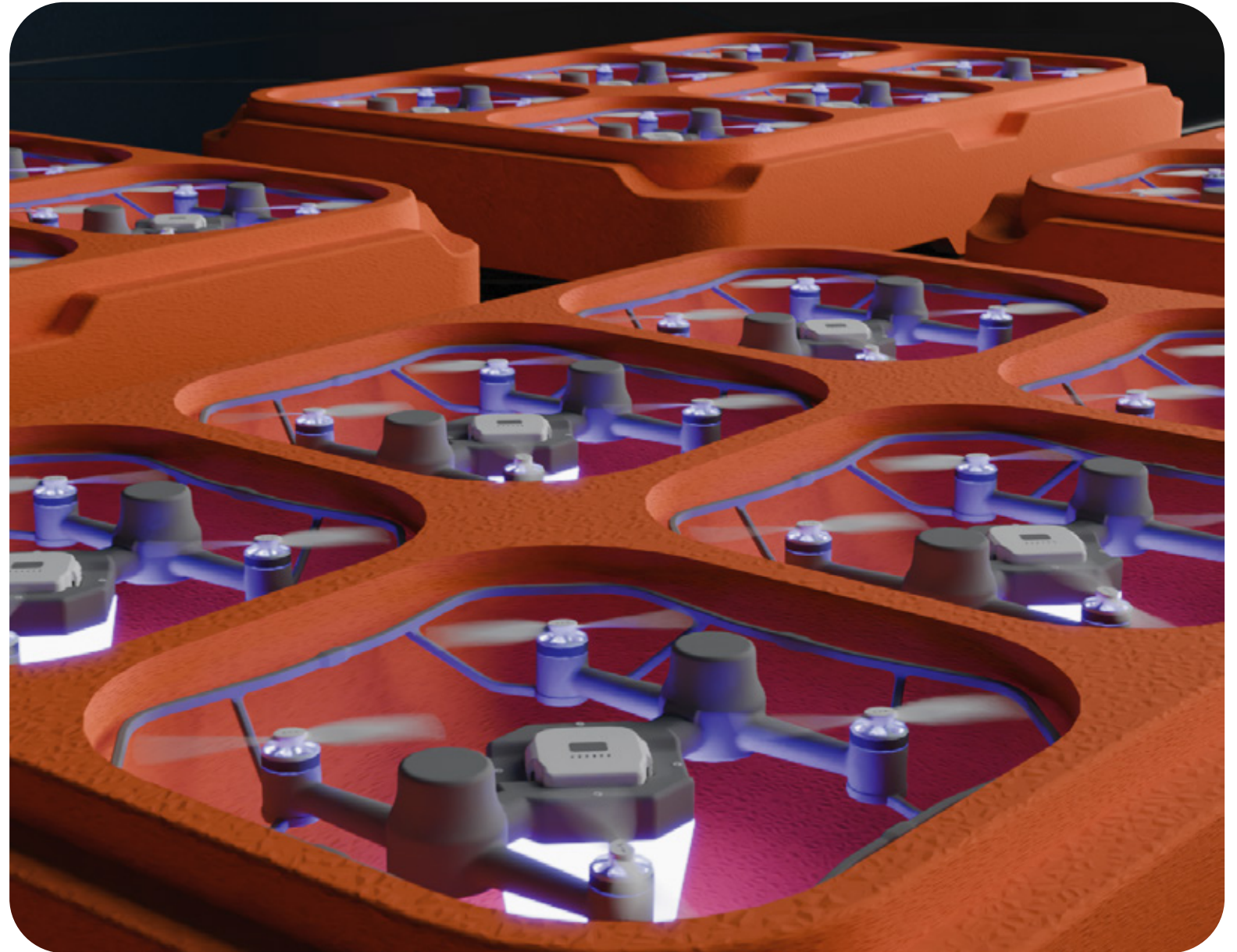
Designed to redefine efficiency, precision, and reliability, NOVA combines intelligent deployment, advanced navigation, and next-generation safety systems into a single, fully integrated platform.

Built for the demands of modern large-scale productions, it sets a new standard for how drone fleets are launched, controlled, and experienced in the sky.

Case-Based Takeoff & Landing

Drones take off and land directly from the transport case, eliminating the need for launch pads or manual handling.

This enables faster turnaround times and cleaner, more efficient logistics on site.



Unmatched Versatility

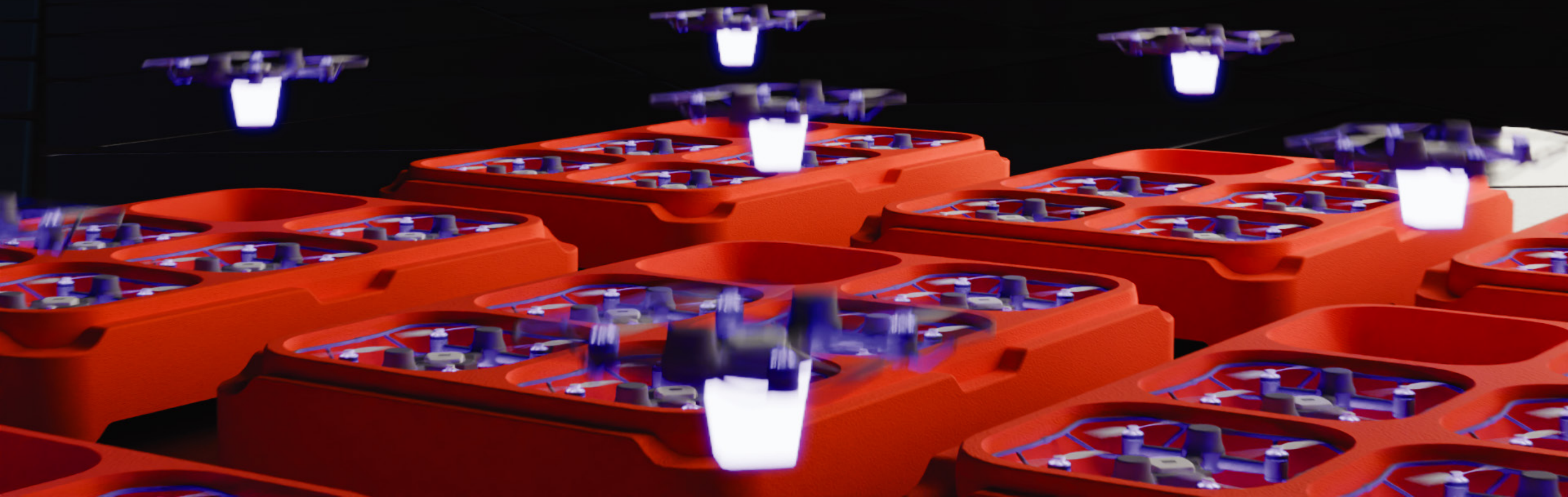
- Case-based takeoff & landing eliminates launch pads
- Fleet launch directly from the case
- Auto Backfill ensures formation continuity

High-Performance Visual System

- 32W RGBW LED system
- Up to 1000m visibility
- Color accuracy for large outdoor environments

Navigation & Safety

- Interference-resistant GNSS
- Dual redundant IMU
- Millimeter-level positioning accuracy
- Rain-Proof design



NOVA PYRO INTEGRATION

Integrated Pyrotechnics – Designed, Not Added

NOVA represents a revolutionary integration in drone show technology. More than an LED platform, NOVA is fully capable of carrying and firing pyrotechnics through an integrated module designed as part of the FLYSTACK ecosystem.

The pyrotechnic legs attach through a fast, secure click-in system, allowing modular configuration without tools or complex setup. Pyrotechnic modules can be fully pre-rigged on the ground and mounted in seconds, significantly reducing preparation time while improving operational safety.

The firing system is completely integrated into the drone, eliminating the need for external firing boxes, additional wiring, or extra equipment.



Click-in pyro legs

No external firing boxes

Altitude-based arming

Automatic disarming on fault

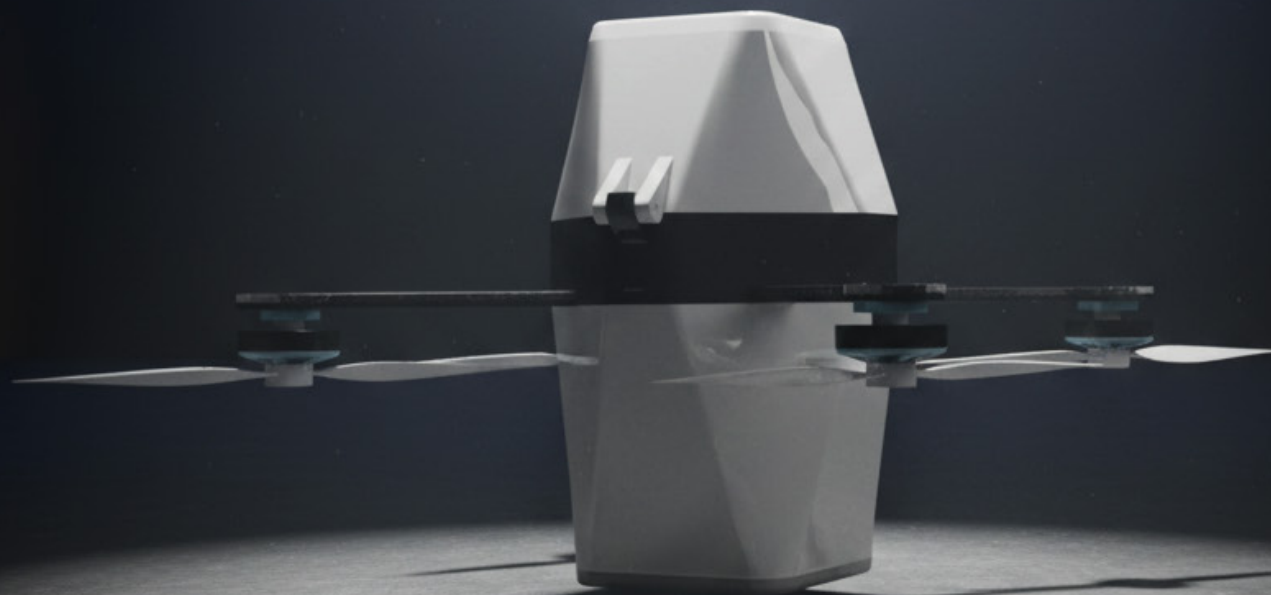
Key Technical Characteristics

Primary Communication	2.4./5.8. GHz (500 m)
Backup Communication	868/915 MHz (2 km)
Communication Protocol	Proprietary
Alternative Protocols	TimeCode, ARTNET, DMX
Control Software	Proprietary
Charger	Proprietary (49 Batteries / Charger)
Configuration Port	USB - C
Magnetometer	Dual
IMU	Dual and Heated
Operating Temperature	-15- 55 °C
Water Resistance	Light Rain Conditions
Takeoff and Landing into the case	Yes
Charging directly into the case	Yes
Side PYRO Shooting	No

Wheelbase	245 mm
Propellers	5 inch
Propellers Guard	Included
Weight	650g
Max Payload	250g
Dimensions	325 x 325 x126mm
LED Solution	RGBW 32 W
Battery	LiPo 5000 mah, 4S, 16.8 V
Pyro Igniters	Optional, 4 channels
Max Speed	10 m/s
Max Flight Time	40 min
Wind Resistance	8 m/s
Positioning System	Multi - Constellation
Flight Termination System	Yes - Triggered by Radio

GP4

GP4 integrates high-performance LED lighting and pyrotechnics into a compact, versatile drone designed for modern aerial entertainment.





GP4

An All-in-One Light & Pyrotechnic Platform

The GP4 represents a new era in aerial entertainment, seamlessly integrating cutting-edge drone technology with both spectacular LED lighting and pyrotechnic capabilities.

This versatile performer delivers the complete package for modern sky shows, eliminating the need for additional systems or control software.

GP4 stands as the industry's most comprehensively integrated drone platform, setting new standards in aerial entertainment technology.

UNMATCHED VERSATILITY

Every aspect of its lightweight frame has been engineered for professional performance: from its dual-redundant sensor systems and super-bright 25W RGBW LED to its multi-layered communication architecture combining 2.4/5GHz WiFi with 868/915MHz backup channel.

The drone achieves remarkable positioning accuracy of 0.8cm through multi-constellation GNSS, while maintaining exceptional magnetic stability even in challenging electromagnetic environments.

What truly sets GP4 apart is how these advanced features - including firework launching capability - seamlessly unite in a single, user-friendly software platform, offering unprecedented control and creative possibilities for aerial displays.

Key Tehnical Specs:

- 25W LED
- multi-constellation GNSS
- redundant systems
- unified software control

Built In Pyrotechnics:

- 3 ch independently triggered
- 150g max payload
- Remote arming/firing system
- Safety integrates features (low altitude disarm, drone faults disarming)
- Dropping effects (waterfall / dropping comets)

Batteries & Chargers:

- 49 batteries capacity
- 1500 W power
- 20 kg weight
- Flight Case



Key Technical Characteristics

Primary Communication	2.4./5.8. GHz (500 m)
Backup Communication	868/915 MHz (2 km)
Communication Protocol	Proprietary
Alternative Protocols	TimeCode, ARTNET, DMX
Control Software	Proprietary - FLYSTACK PILOT
Charger	Proprietary (49 Batteries / Charger)
Configuration Port	USB - C
Magnetometer	Dual
IMU	Dual and Heated
Operating Temperature	-15 - 55 °C
Water Resistance	Light Rain Conditions
Takeoff and Landing into the case	No
Charging directly into the case	No
Side PYRO Shooting	No

Wheelbase	220 mm
Propellers	5.1 inch
Propellers Guard	optional
Weight	630g
Max Payload	150g
Dimensions	310 x 310 x 140 mm
LED Solution	RGBW 32 W
Battery	LiPo 3500 mah, 4S, 17.2 V
Pyro Igniters	3 channels, quick release
Max Speed	10 m/s
Max Flight Time	25 min
Wind Resistance	8 m/s
Positioning System	Multi - Constellation
Flight Termination System	Yes - Triggered by Radio



MD6

The MD6 represents a new era in aerial entertainment, seamlessly integrating cutting-edge drone technology with both spectacular LED lighting and pyrotechnic capabilities.

MD6

MD6 stands as the industry's most comprehensively integrated drone platform, setting new standards in aerial entertainment technology.

Every aspect of its lightweight frame has been engineered for professional performance: from its dual-redundant sensor systems and super-bright 32W RGBW LED to its multi-layered communication architecture combining 5GHz WiFi with 868/915MHz backup channel.

The drone achieves remarkable positioning accuracy of 0.8cm through multi-constellation GNSS, while maintaining exceptional magnetic stability even in challenging electromagnetic environments.





High-Payload, High-Impact Aerial Performer

What truly sets MD6 apart is how these advanced features - including firework launching capability – seamlessly unite in a single, user-friendly software platform, offering unprecedented control and creative possibilities for aerial displays.

Purpose-built for professional aerial displays, the MD6 combines a robust 32W RGBW LED system with pyrotechnic carrying capability in a compact 950g package.

With up to 20 minutes of flight time, it offers show designers unprecedented creative freedom to blend light and fire in the night sky.

Key Tehnical Specs:

- 950 g takeoff weight
- Up to 20 min flight time
- 32 W RGBW LED
- 5.8 GHz communication
- Double redundant IMU & magnetometer
- 868 MHz backup control
- 0.8 cm positioning accuracy
- 9.5 inch propellers
- Remote power ON/OFF

Built In Pyrotechnics:

- 6 ch independently triggered
- 500 g max payload
- Remote arming system
- Dropping effects (waterfall / dropping comets)
- Side shorting effects (single shots / comets / mines)

Batteries & Chargers:

- 49 batteries capacity
- 2500 W power
- 20 kg weight
- Flight Case



Key Technical Characteristics

Primary Communication	2.4. / 5.8. GHz (500 m)
Backup Communication	868 / 915 MHz (2 km)
Communication Protocol	Proprietary
Alternative Protocols	TimeCode, ARTNET, DMX
Control Software	Proprietary
Charger	Proprietary (49 Batteries / Charger)
Configuration Port	USB - C
Magnetometer	Dual
IMU	Dual and Heated
Operating Temperature	-15 - 55 °C
Water Resistance	Heavy Rain Conditions
Takeoff and Landing into the case	No
Charging directly into the case	No
Side PYRO Shooting	Yes

Wheelbase	495 mm
Propellers	9.5 inch
Propellers Guard	N/A
Weight	1300g
Max Payload	500g
Dimensions	470 x 470 x 210 mm
LED Solution	RGBW 32 W
Battery	LiPo 5500 mah, 4S, 17.2 V
Pyro Igniters	6 channels
Max Speed	10 m/s
Max Flight Time	20 min
Wind Resistance	15 m/s
Positioning System	Multi - Constellation
Flight Termination System	Yes - Triggered by Radio

MD12

MD12 is a heavy-duty pyrotechnic drone designed for high-impact, large-scale aerial fire performances.





MD12

Built from the ground up for certified aerial fireworks, MD12 combines high payload capacity with advanced flight stability, making it a reliable platform for complex pyrotechnic sequences, dynamic fire effects, and synchronized light & fire shows.

Its robust airframe, industrial-grade components, and reinforced propulsion system allow safe operation even under demanding conditions and strong wind environments.

At the core of MD12 lies a fully integrated multi-channel pyrotechnic firing system, capable of independently triggering multiple effects without the need for external firing boxes or additional wiring.

Heavy-Duty Pyrotechnic Drone for Certified Aerial Fireworks

Advanced redundancy systems, including dual IMU and dual magnetometer architecture, ensure precise control and stable positioning throughout the flight.

High-accuracy multi-constellation navigation enables tight formations and repeatable choreography, even in complex RF or urban environments.

Designed for professional show workflows, MD12 integrates seamlessly into the FLYSTACK ecosystem, supporting proprietary control software as well as industry-standard protocols such as timecode, Art-Net, and DMX.



UNMATCHED VERSATILITY

Key Tehnical Specs:

- 1250 g takeoff weight
- Up to 22 min flight time
- 32 W RGBW LED
- 5.8 GHz communication
- Double redundant IMU & magnetometer
- 868 MHz backup control
- 0.8 cm positioning accuracy
- 9.5 inch folding propellers
- Remote power ON/OFF

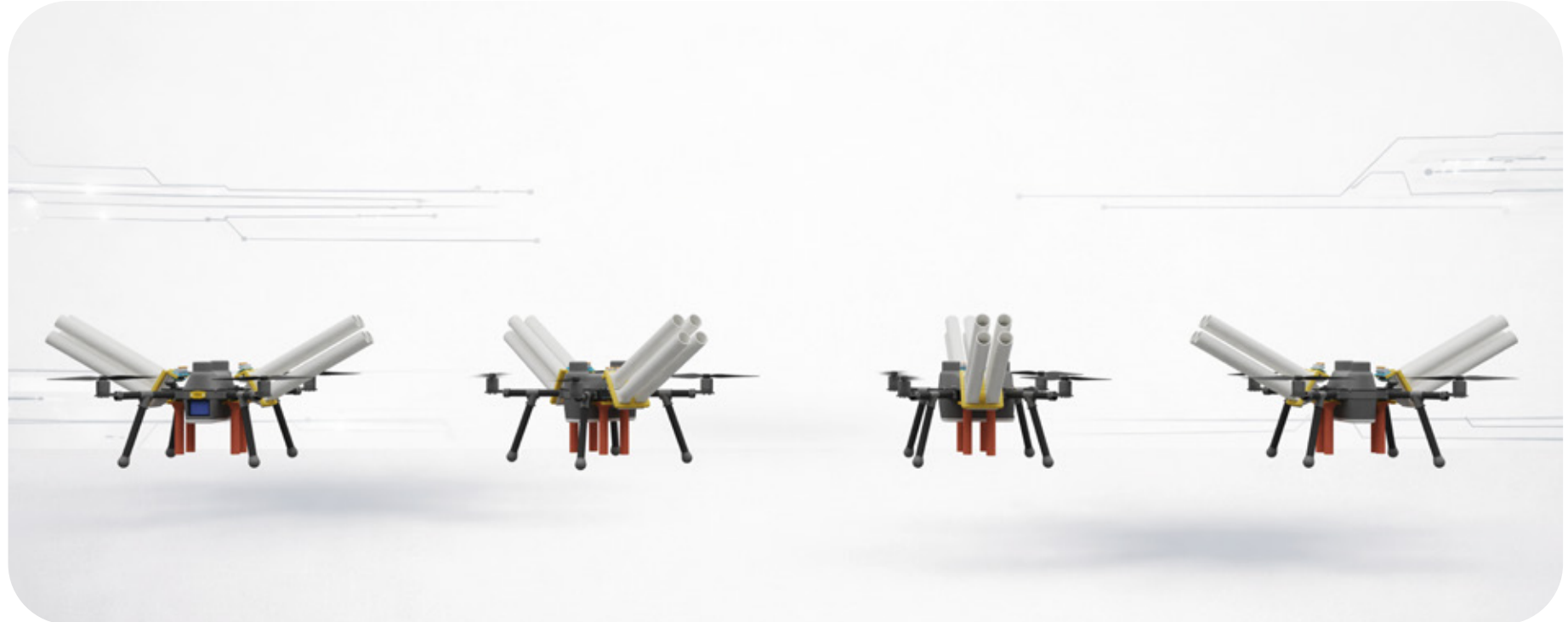
Built In Pyrotechnics:

- 12 ch independently triggered
- 750 g max payload
- Remote arming/firing system
- Dropping effects (waterfall / dropping comets)
- Side shorting effects (single shots / comets / mines)

Batteries & Chargers:

- 49 batteries capacity
- 2500 W power
- 20 kg weight
- Flight Case

With high weather resistance, remote arming and firing, and support for advanced dropping and side-shooting effects, MD12 delivers a powerful, scalable solution for modern aerial pyrotechnics - where safety, precision, and visual impact are non-negotiable.



Key Technical Characteristics

Primary Communication	2.4./ 5.8. GHz (500 m)
Backup Communication	868 / 915 MHz (2 km)
Communication Protocol	Proprietary
Alternative Protocols	TimeCode, ARTNET, DMX
Control Software	Proprietary
Charger	Proprietary (49 Batteries / Charger)
Configuration Port	USB - C
Magnetometer	Dual
IMU	Dual and Heated
Operating Temperature	-15 - 55 °C
Water Resistance	Heavy Rain Conditions
Takeoff and Landing into the case	No
Charging directly into the case	No
Side PYRO Shooting	Yes

Wheelbase	400 mm
Propellers	9.5 inch
Propellers Guard	N/A
Weight	1500g
Max Payload	750g
Dimensions	520 x 520 x 200 mm
LED Solution	RGBW 32 W
Battery	LiPo 5500 mah, 4S, 17.2 V
Pyro Igniters	12 channels
Max Speed	10 m/s
Max Flight Time	20 min
Wind Resistance	15 m/s
Positioning System	Multi - Constellation
Flight Termination System	Yes - Triggered by Radio



CONTROL SOFTWARE

Software as the Backbone of the Show

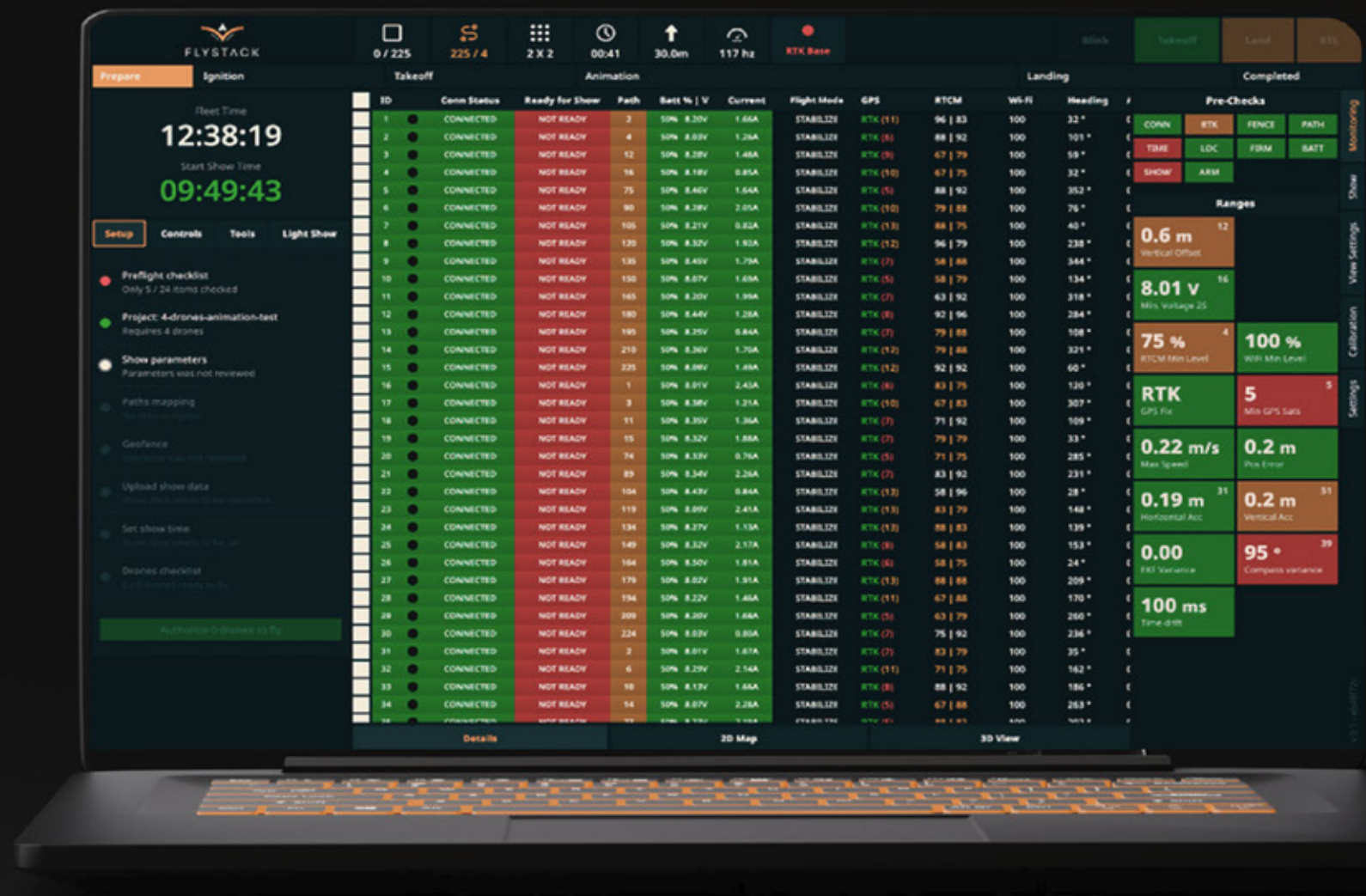
PILOT

At the heart of our drones's performance lies our proprietary control software, purpose-built for professional drone entertainment.

The system seamlessly integrates with industry-standard protocols including TimeCode and ARTNET, enabling perfect synchronization with existing show elements.

While automation handles complex choreography, pilots maintain full manual control and real-time monitoring capabilities through an intuitive interface that delivers critical flight data and alerts instantly.

Multiple geofence layers and comprehensive safety protocols work in concert to ensure secure operations, making risk management an integral part of every performance.



CORE CAPABILITIES

PILOT

Fleet Management

- Real-time telemetry across the full fleet (GPS, voltage, RSSI, health, temperature, ESC data)
- Automatic identification + serial tracking for inventory and deployment history
- Multi-model configuration profiles + companion-computer diagnostics & sync

Show Planning & Execution

- 2D/3D monitoring and realistic preview modes
- Automatic drone-to-path mapping + grid-based takeoff layout
- Timeline-based sequencing with precise GPS-time start

Safety & Resilience

- Multi-layer containment: Soft Fence (warning polygon) + Hard Fence (failsafe polygon) + Ballistic Fence (predicted impact / hit zone)
- If the predicted hit zone exceeds a defined threshold, the system can trigger emergency flight termination logic (motor disable) as a last-resort protection measure
- Redundant comms with auto-failover + network topology and signal quality indicators
- Heartbeat supervision for flight controller + companion systems, configurable thresholds

RTK & Redundant Commanding

- RTK integration for precise positioning and stable formation geometry
- Primary + secondary command channel, with commands sent from the same or a separate

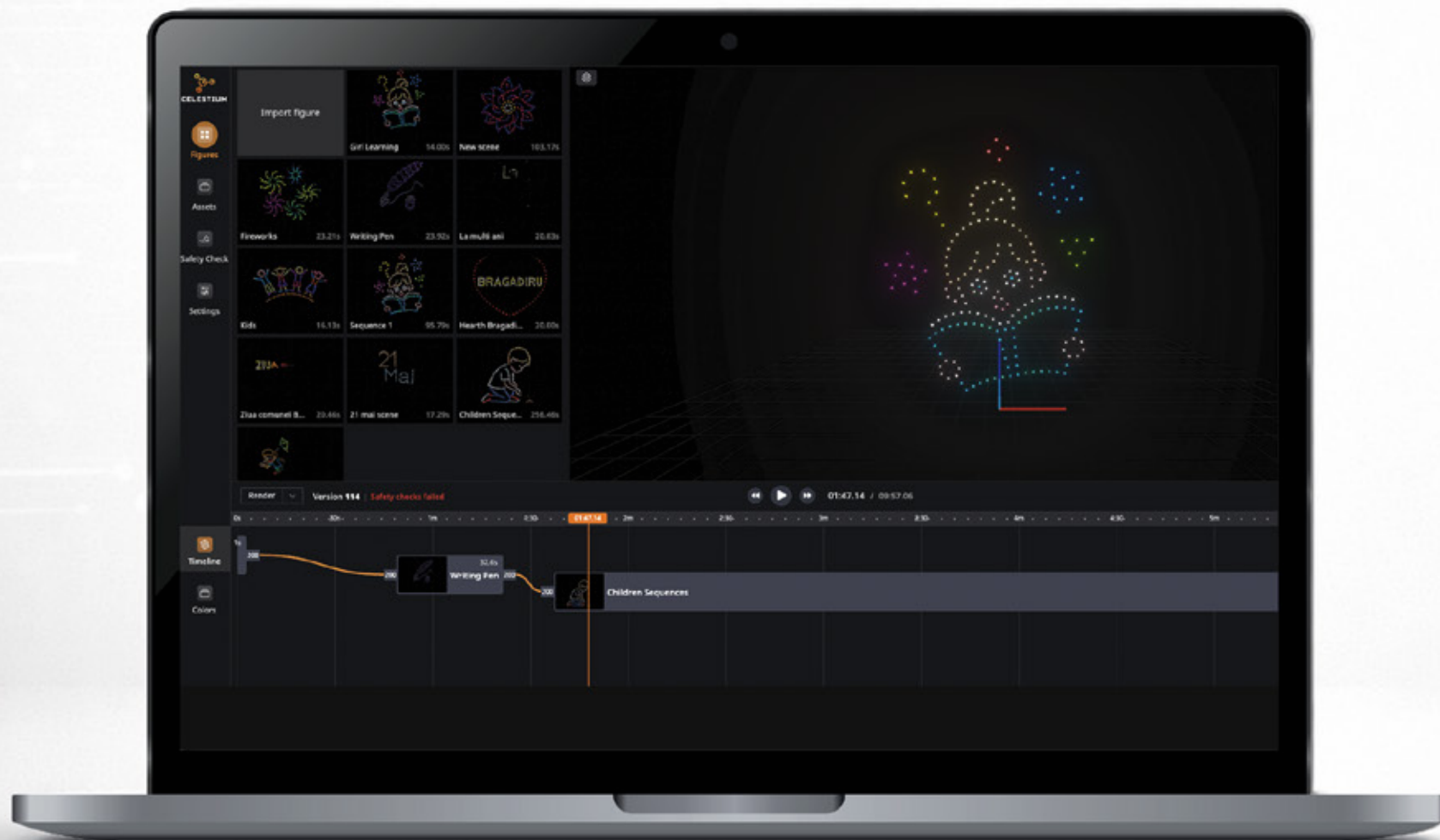
Reporting & Compliance

- Post-flight metrics per drone, anomaly detection, aggregated show stats
- Exportable reports for documentation and regulatory requirements

Preflight Automation & Validation

- Automated drone discovery, network assignment, file distribution, and parameter consistency checks
- Automatic pre-flight checks with clear go/no-go status across the fleet (comms, GPS/RTK, power, sensors, health)
- Show file verification to detect collision risks and speed/acceleration violations before launch





CELESTIUM

CELESTIUM empowers creators of all levels to transform their ideas into unforgettable drone light shows. With an intuitive interface, real-time previews, and powerful animation tools, it eliminates technical barriers and streamlines the creative process from start to finish.

CORE CAPABILITIES

Designed for All Creators:

Whether you're a professional animator, a show operator, or just getting started, CELESTIUM is built to support you. No prior 3D design experience is required. Our userfriendly environment allows you to focus on what matters most: creativity.

User-Friendly Interface:

- Clean and accessible workspace designed for efficiency
- Intuitive top menu structure (File, Edit, View, Build)
- Quick-access shortcuts and customizable panels
- Drag-and-drop workflow for smooth navigation

Custom Asset Libraries:

- Import and organize your files into structured libraries
- Store images, videos, sounds, fonts, and 3D models in dedicated folders
- Quickly add new assets via right-click or drag-and-drop
- Optimize your workflow by reusing materials across projects

Efficient Node-Based Editor:

- Build animations by connecting modular nodes
- Modify node properties instantly with real-time feedback
- Link visual, audio, or movement-based elements for complex sequences
- Reusable components streamline creation and improve consistency

Flexible Drone Control:

- Manipulate individual drones or entire groups in the Scene 3D view
- Choose between global and local positioning for detailed control
- Real-time adjustments visible instantly in the visualizer
- Precision tools for trajectory, timing, speed, and formation control

Import and Export Projects:

- Save and load full show projects including all assets
- Collaborate easily by sharing packaged show files
- Use templates or adapt previous work to speed up production

3D Model Integration

- Support for popular 3D formats;
- Import, scale, and position models in the scene seamlessly
- Integrate logos, architecture, or artistic visuals directly into the animation

Text & Audience Interaction:

- Create text nodes with customizable fonts, styles, and animations
- Display messages, names, countdowns, or dynamic titles that enhance the narrative
- Combine text with synchronized visuals to boost audience engagement and storytelling impact

Image & Video Projections:

- Map visuals directly onto the drone formations
- Adjust projection height, rotation, and scale
- Synchronize dynamic content with choreography for immersive storytelling

Real-Time Preview and Fine-Tuning:

- All changes visible instantly in the 3D viewer
- Preview the entire show or specific sequences
- Ensure perfect timing and alignment before deployment



CELESTIUM

From Vision to Show Faster Than Ever

CELESTIUM significantly reduces the time from concept to execution. Whether you're working on a small local show or a massive multimedia production, our tools are designed to help you deliver powerful, polished results—faster than ever.





FLYSTACK SERVICES

Drone Show Design & Production



01. Drone Show Design & Production

We provide complete, end-to-end drone show design and production services, transforming ideas into fully synchronized aerial performances.

From the initial creative concept and visual storytelling to technical planning, animation, and live execution, every stage is developed in-house to ensure precision, safety, and consistency.

Our creative teams work closely with clients to craft original narratives, symbols, and transitions that deliver strong emotional and visual impact.

On the technical side, we manage flight choreography, positioning accuracy, redundancy planning, and full synchronization with music, timecode, lighting, lasers, fireworks, and multimedia systems.

Using our proprietary control and animation software, we ensure smooth transitions, tight formations, and reliable execution, even in complex environments or large-scale productions involving hundreds or thousands of drones.

Our production services also include risk assessment, aviation compliance, safety documentation, and on-site operational management.

With extensive international experience and a proven technology stack, we deliver drone shows that meet the highest professional standards—scalable, repeatable, and designed to perform flawlessly in real-world conditions.





02. Creative Live Concept Development

We create original live concepts that go beyond standalone drone shows, designing immersive experiences where technology, storytelling, and emotion come together. Our creative process starts with a clear narrative vision, transforming abstract ideas, brand messages, or cultural themes into dynamic live moments designed specifically for large-scale audiences.

By combining drones with pyrotechnics, lighting, lasers, video mapping, water projections, and synchronized sound, we develop cohesive multimedia concepts that unfold seamlessly in real time. Every element is designed to work in perfect harmony, both visually and technically, through precise timecode-based synchronization.

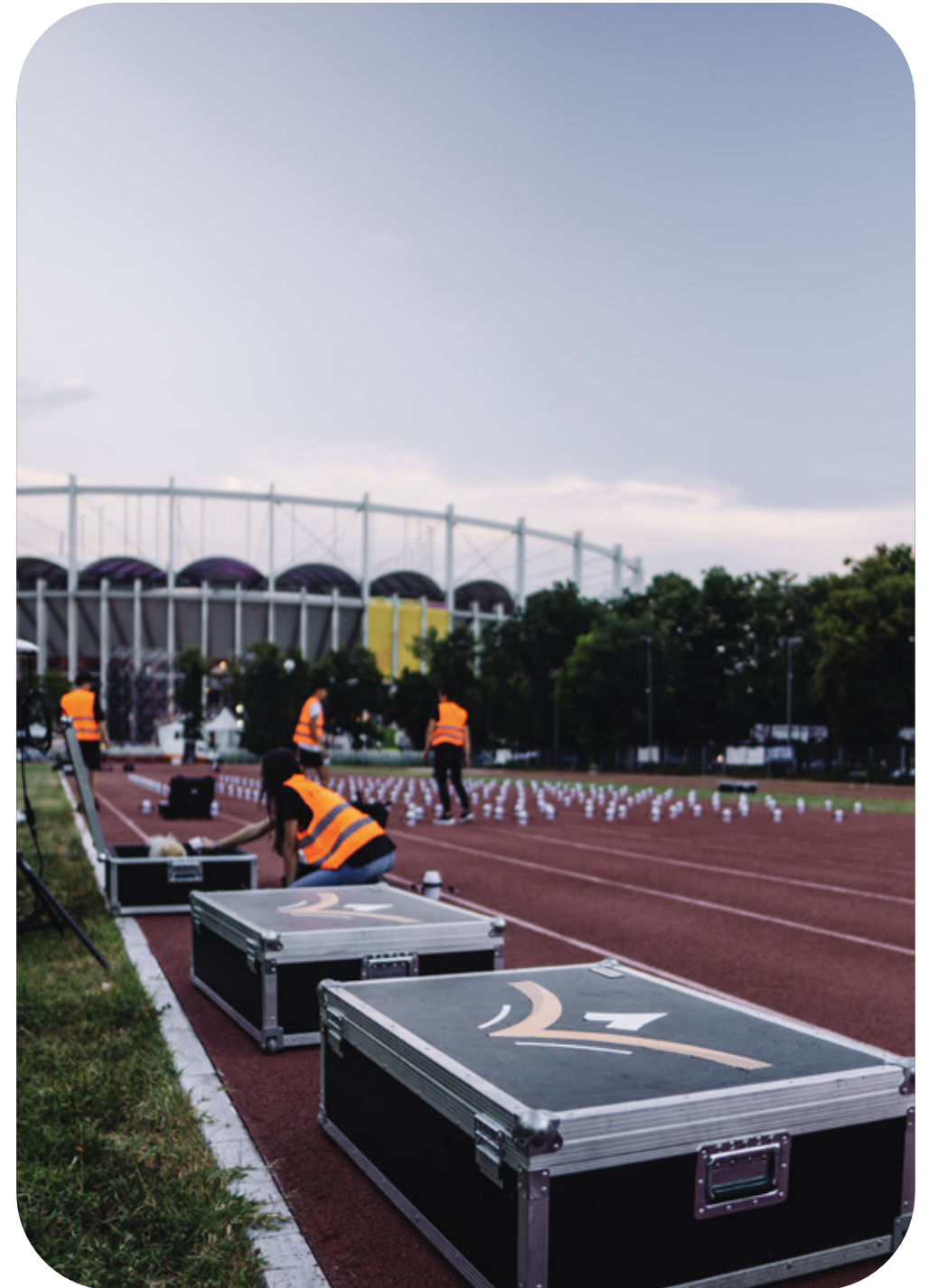
From concept ideation and visual design to technical feasibility and live execution planning, we ensure that each creative concept is not only visually powerful, but also fully achievable, safe, and scalable. The result is a unique live experience—engineered for impact, reliability, and memorability.

03. Training, Consulting, System Deployment & Authorisation Support

We provide comprehensive training, consulting, and system deployment services for professional drone show operations worldwide. Our programs are designed for pilots, technical teams, and operators, covering flight operations, safety procedures, software usage, system maintenance, and live show workflows. Training can be delivered on-site or remotely, tailored to the experience level and operational needs of each team.

In parallel, we offer technical and strategic consulting for drone show projects, supporting system configuration, operational planning, and large-scale deployment. Our team assists with the setup and integration of hardware and software into existing production environments, ensuring optimal performance and reliability.

We also provide full support for aviation authorisations and regulatory compliance, including risk assessments, safety documentation, operational manuals, and coordination with local aviation authorities. With experience across multiple international markets, we help ensure that every operation meets legal, technical, and safety requirements—allowing teams to focus on creative execution with confidence.



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FLYSTACK